

CHE 205

1. Course number and name

CHE 205. Computational Methods in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Shafigh Mehraeen, Betul Bilgin

4. Textbook, title, author, publisher, year

Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law.
ISBN: 978-1-4665-7534-9, CRC Press, 2013.

a. Other supplemental materials

VBA programming; tutorials on the internet.

Consult with the following websites:

[http://excelvbatutor.com/vba tutorial.html](http://excelvbatutor.com/vba%20tutorial.html)

[http://www.homeandlearn.org/basic mathematical operators.html](http://www.homeandlearn.org/basic%20mathematical%20operators.html)

[http://www.automateexcel.com/vba-code-examples.](http://www.automateexcel.com/vba-code-examples)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Computational methods and software relevant to unit operations. Excel spreadsheets (curve fitting, heat conduction), Matlab, Aspen Plus (process simulation), algorithms and object-oriented concepts in chemical engineering.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 201 and MATH 210.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Formulate a chemical engineering problem as a mathematical model and select appropriate solution method.
- Write and modify VBA scripts for specific computational purposes and run them.
- Use Excel functionality to tabulate thermophysical property correlations and equations of state.
- Analyze 1d (linear) and 2d (bilinear) interpolation from data tables with VBA macros.
- Calculate vector coordinates for changes of basis vectors with Excel.
- Solve ordinary differential equations (ODE) with Excel.
- Solve optimization design problems with Excel Goal Seek and Solver functionalities and VBA macros.

- Solve partial differential equations (PDE).
- Carry out error analysis of data and regression analysis.
- Find root of a function using bisection, secant, fixed-point iteration, and Newton's method.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		

7. Brief list of topics to be covered

- Ch. 1: Roots of a single nonlinear equation
- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver